Ilaria Trombini

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Scopus Research gate Google Scholar

Personal Details

Born in 1997 in Portomaggiore (Italy). Italian citizen.
Languages: English (fluent), Italian (mother tongue)

Current position

Post-Doctoral researcher (October 2024–present), University of Ferrara, Italy.

Education and degrees completed

PhD	**University of Parma**, Mathematical, Physical and Computer Sciences	21 March 2025
MS	**University of Ferrara**, Mathematics and Computer Science	27 July 2021
BS	**University of Ferrara**, Mathematics and Computer Science	26 July 2019

Research interests

Stochastic gradient-like methods for machine and deep learning applications and inverse problems: investigation about adaptive steplength selection rules, line-search procedures, reductions variance techniques by dynamic adaptive mini-batch size.

Previous work experience

University of Ferrara, Civil and Environmental, Electrical and Computer science, Mechanical Engineering. Supplementary lecturer on first-semester coursework (25 hours) — Sep 2026 - Nov 2026

University of Ferrara, Mathematics and Computer Science. Co-Lecturer (8/56 hours, 1 CFU) of Analisi Numerica I for the Bachelor's Degree in Mathematics — November 2025

University of Ferrara, Mathematics and Computer Science. Lecturer on tensor calculus (8 hours) — September 2024

University of Ferrara, Mathematics and Computer Science. Lecturer on tensor calculus (8 hours) — September 2023

University of Ferrara, Mathematics and Computer Science. Lecturer on tensor calculus (8 hours) — October 2022

Meeo s.r.l, Internship — Apr 2021 – May 2021

- Study of the development of the locust and predictions of its movement

University of Ferrara, Mathematics and Computer Science. Tutoring of Calcolo numerico e laboratorio (27 hours) — Sep 2021 – Jan 2024

University of Ferrara, Civil engineering. Tutoring of Introduzione a Matlab (25 hours) — Sep 2022 – Nov 2023

University of Ferrara, Mathematics and Computer Science. Tutoring Progetto Lauree Scientifiche (76 hours) — Nov 2021 – Mar 2022

University of Ferrara, Mathematics and Computer Science. Tutoring of Analisi Numerica I (24 hours) — Oct 2019 - Jan 2025

Personal research funding and awards

Awards

Honorable Mentions for “Daniela di Serafino” International Doctoral Award - 2025 — 2025

Funding as PI

INdAM - GNCS Projects 2026, "Oltre il gradiente deterministico: analisi di dinamiche stocastiche per funzioni non convesse". Amount: 1800 euros — 2026

Bando di selezione “Universities for Innovation” - 35 Borse di mobilità Erasmus+ ai fini di formazione per lo staff degli istituti si istruzione superiori. Amount: 3073 euros. — 2026

5x1000 - Dipartimento di Matematica ed Informatica, Unife, "Algoritmi stocastici di ottimizzazione applicati all’ambito dei problemi inversi". Amount: 1990 euros. — 2025

18a edizione Borse di Studio Favlaf a favore di studenti Unife figli di operai agricoli. Amount: 1100 euros. — 2018

17a edizione Borse di Studio Favlaf a favore di studenti Unife figli di operai agricoli. Amount: 1100 euros. — 2017

Participation to research projects

5x1000 - Dipartimento di Matematica ed Informatica, Unife, "Metodi particellari multiscala e apprendimento automatico per operatori collisionali in equazioni cinetiche". (PI: A. Medaglia) — 2026

INdAM - GNCS Projects 2025, "Metodi avanzati di ottimizzazione stocastica per la risoluzione di problemi inversi di imaging". (PI: T. A. Bubba) — 2025

Advanced MATHematical methods for Artificial Intelligence – MATH4AI, funded by the programma "Future Artificiale Intelligence, FAIR” PE0000013, CUP DJ33C22002830006 – Mission 4 Component 2, Investment 1.3 (MUR Directorial Decree no. 341 of 03/15/2022) — 2024

INdAM - GNCS Projects 2024, "Deep Variational Learning: a combined approach for image reconstruction". (PI: A. Benfenati) — 2024

INdAM - GNCS Projects 2023, "Data-driven optimization methods: new theoretical and practical perspectives". (PI: G. Franchini) — 2023

INdAM - GNCS Projects 2022, "Adaptive optimization for machine learning". (PI: S. Rebegoldi) — 2022

List of publications

Submitted

1. N. Krklec Jerinkić, L. Rutešić, and I. Trombini. ASMOP: Additional sampling stochastic trust region method for multi-objective problems. (2026).
2. T. A. Bubba, E. Morotti, F. Porta, V. Ruggiero and I. Trombini. A Line–Search–Based Stochastic Gradient Method for 3D Computed Tomography. (2026).
3. M. Español, M. Juvonen, S. Siltanen, I. Trombini and C. Wu. A Structured-Surrogate Double-Model Approach to Regularization Parameter Selection. (2026).

Journal article

1. N. Krklec Jerinkić, F. Porta, V. Ruggiero, and I. Trombini. Variable metric proximal stochastic gradient methods with additional sampling. In: Comput Optim Appl (2025).
2. N. Krklec Jerinkić, V. Ruggiero, and I. Trombini. Spectral Stochastic Gradient Method with Additional Sampling for Finite and Infinite Sums. In: Comput Optim Appl 91 (2025), pp. 717–758.

3. G. Franchini, F. Porta, V. Ruggiero, I. Trombini, and Zanni L. A stochastic gradient method with variance control and variable learning rate for Deep Learning. In: Journal of Computational and Applied Mathematics 451 (2024), p. 116083.
4. G. Franchini, F. Porta, V. Ruggiero, and I. Trombini. Correction to: A Line Search Based Proximal Stochastic Gradient Algorithm with Dynamical Variance Reduction. In: Journal of Scientific Computing 96.48 (2023).
5. G. Franchini, F. Porta, V. Ruggiero, I. Trombini, and L. Zanni. Learning rate selection in stochastic gradient methods based on line search strategies. In: Applied Mathematics in Science and Engineering 31.1 (2023), p. 2164000.
6. G. Franchini, F. Porta, V. Ruggiero, and I. Trombini. A line search based proximal stochastic gradient algorithm with dynamical variance reduction. In: Journal of Scientific Computing 94.23 (2023).

Conference proceedings

1. G. Franchini, F. Porta, V. Ruggiero, I. Trombini, and L. Zanni. Line Search Stochastic Gradient Algorithm with a-priori rule for monitoring the control of the variance. In: Sergeyev, Y.D., Kvasov, D.E., Astorino, A. (eds) Numerical Computations: Theory and Algorithms. NUMTA 2023. Lecture Notes in Computer Science. Springer, Cham. 14476 (2025).
2. G. Franchini, F. Porta, V. Ruggiero, I. Trombini, and L. Zanni. Diagonal Barzilai-Borwein Rules in Stochastic Gradient-Like Methods. In: Dorronsoro, B., Chicano, F., Danoy, G., Talbi, EG. (eds) Optimization and Learning. OLA 2023. Communications in Computer and Information Science. Vol. 1824. (2023), pp. 21–35.
3. G. Franchini, V. Ruggiero, and I. Trombini. Thresholding Procedure via Barzilai-Borwein Rules for the Steplength Selection in Stochastic Gradient Methods. In: Nicosia, G., et al. Machine Learning, Optimization, and Data Science. LOD 2021. Lecture Notes in Computer Science 13164 (2022).

Thesis

1. I. Trombini. On the hyperparameters setting for first order stochastic optimization methods in machine learning. PhD Thesis. (2025).

Experience of organising scientific meetings

Organizer with G. Malaspina of Minisymposium "Optimization in Action: Advances in Stochastic and Deterministic Methods" **Young Applied Mathematicians Conference YAMC2025**, Padova — September 2025

Member of the Scientific Committee: **Two-days follow on Advanced Numerical Methods for Machine** & **Deep Learning**, Ferrara — February 2025

Referee activity for scientific journals

Reviewer for Mathematical Reviews.

Society memberships

Member of European Women in Mathematics (**EWM**) — 2026-Present

Member of UMI - Matematica per l'intelligenza artificiale e il machine learning (**AI**&**ML**&**MAT**) — 2025-Present

Member of UMI - Matematica delle Immagini, della Visione e delle loro Applicazioni (**MIVA**) — 2025-Present

Member of Unione Matematica Italiana (**UMI**) — 2025-Present

Member of Optimization Algorithms and Software for Inverse Problems (**OASIS**) — 2021-Present

Member of National Group of Scientific Computing (**GNCS**) — 2021-Present

Significant administrative responsibilities

Student representative in the CdS and CdD of the Department of Mathematics and Computer Science at University of Ferrara.

Research visits to other Institutions

In years 2021-2026, **2 extended visits** (2 months in Spring 2023 at University of Novi Sad, Serbia; 3 week in Spring 2026 at University of Helsinki, Finland) and **2 short visit** (2 days in Spring 2026 at Centro comune di ricerca (Commissione europea) | JRC Ispra, Italy; 2 days in Autumn 2026 at Centro comune di ricerca (Commissione europea) | JRC Ispra, Italy).

Supervision and Co-supervision

Thesis co-supervision of "Applicazione di metodi di discesa del gradiente al problema inverso della sfocatura di immagini", Fois Matteo (**first level degree in Mathematics**)	2025-2026
Thesis secondary supervisor of "Il problema inverso della ricostruzione di immagini sfocate", Carra Irene (**first level degree in Mathematics**)	2023-2024
Thesis secondary supervisor of "Metodi numerici per il calcolo di autovalori e autovettori", Fabbri Anna (**first level degree in Mathematics**)	2022-2023
Thesis secondary supervisor of "Addestramento automatico mediante regressione logistica e SVM", Bignozzi Dario (**first level degree in Mathematics**)	2021-2022

Scientific talks

Conference talks

Conference "**Conference Young Applied Mathematicians (YAMC2026)**". Polytechnic of Turin (invited talk)	Sep 2026
Workshop "**Numerical Optimization: Theory, Algorithms, Applications**". University of Modena and Reggio Emilia (invited talk)	Sep 2026
Conference "**Conference on Advances in Continuous Optimization (EUROPT 2026)**". Linz (invited talk)	Jul 2026
Conference "**GIMC SIMAI YOUNG 2026**". Pisa (invited talk)	Jun 2026
Conference "**SIAM Chapters meeting for Young Researchers in Applied Mathematics and Scientific Computing**". PoliMi	Jan 2026
Conference "**Applied Inverse Problems (AIP 2025)**". FGV EMAp, Rio de Janeiro (invited talk)	Jul 2025
Conference "**International Conference On Continuous Optimization (ICOOPT 2025)**". University of Southern California (invited talk)	Jul 2025
Workshop "**Mathematics for Artificial Intelligence and Machine Learning**". University of Bari	Jan 2025
Conference "**25th International Symposium on Mathematical Programming (ISMP 2024)**". Montrèal (invited talk)	Jul 2024
Workshop "**GIMC SIMAI YOUNG 2024**". University of Napoli (invited talk)	Jul 2024
Workshop "**Algorithms' Impact on Artificial Intelligence (AiAi)**". University of Bari	Jun 2024
Conference "**Conference Young Applied Mathematicians (YAMC)**". University of Siena	Sep 2023
Conference "**Conference numerical computations: theory and algorithms (NUMTA2023)**". University of Calabria (invited talk)	Jun 2023

Workshop "**The mathematics of machine learning**". University of Pisa - Centro de Giorgi (invited talk) — Jan 2023

Poster session

Workshop "**MAthematical and Computational imaging acROss modalities (MACRO)** ". INdAM Workshop, Rome — Jun 2026

Workshop "**Italian Inverse problems** & **Imaging meeting** ". University of Genoa — Jan 2026

Workshop "**Annual Meeting of EMS activity group on Scientific Machine Learning**". University of Milano — Mar 2025

Workshop "**Optimization Techniques for Inverse Problems (OIP 2024)**". University of Modena — Sep 2024

Public outreach

- Representative at the Mathematics booth during “European Researchers’ Night”.
- Representative at the Mathematics booth during “Unife Orienta”.
- Representative at the Mathematics booth during “Porte Aperte al Polo Scientifico Tecnologico”.
- Teacher for the mathematics internship reserved for 4th year high school classes.